# MATH 110 — Week 6

## Homework 6 — Solutions

**HW6 P1.**

**Idea.** The equation $P^2 = P$ controls eigenvalues and supplies a decomposition for every vector.

1. Assume the intended linearity hypothesis. If $Pv = \lambda v$ with $v \neq 0$, then $P^2 v = \lambda^2 v$, while $P^2 v = Pv = \lambda v$. Hence $\lambda(\lambda - 1) = 0$, so $\lambda \in \{0, 1\}$. It is not asserted that both occur.
2. Every $v = (v - Pv) + Pv$, with $P(v - Pv) = Pv - P^2 v = 0$ and $Pv \in \operatorname{range} P$. Thus $V = \ker P + \operatorname{range} P$.
3. If $w \in \ker P \cap \operatorname{range} P$, write $w = Pu$. Then $w = Pu = P^2 u = Pw = 0$, so the sum is direct.

**HW6 P2(a).**

**Idea.** Differentiation lowers the degree of every nonconstant polynomial over $\mathbb{R}$ or $\mathbb{C}$.

1. If $Dp = \lambda p$ and $p \neq 0$ with $\lambda \neq 0$, a nonconstant $p$ would give degree $\deg p - 1$ on the left and $\deg p$ on the right, a contradiction. A nonzero constant gives zero on the left and nonzero on the right, also impossible.
2. The eigenvalue 0 occurs because $D1 = 0$. Writing $p = \sum_{j=0}^{n} a_j x^j$ gives $Dp = \sum_{j=1}^{n} j a_j x^{j-1}$. A zero derivative forces each $j a_j = 0$, hence $a_j = 0$ for $j \geq 1$ in either field. Thus the kernel consists of constants.
3. Thus the only eigenvalue is 0 and $E(0, D) = \operatorname{span}(1)$; for $\lambda \neq 0$, $E(\lambda, D) = \{0\}$. This also covers $n = 0$.

**HW6 P2(b).**

**Idea.** Smooth functions include nonpolynomial exponential solutions of the derivative equation.

1. We cite elementary calculus: exponential differentiation, the product rule, and the fact that a function with derivative zero on $\mathbb{R}$ is constant. An eigenfunction is a nonzero function satisfying $Df = \lambda f$. On smooth real-valued functions, for every real $\lambda$ the function $e^{\lambda x}$ is a nonzero eigenfunction because its derivative is $\lambda e^{\lambda x}$.
2. If $f' = \lambda f$, then $(e^{-\lambda x} f(x))' = 0$, so $f = c e^{\lambda x}$. Thus $E(\lambda, D) = \operatorname{span}(e^{\lambda x})$.
3. A complex-valued function is smooth when its real and imaginary parts are smooth; differentiation acts on those parts separately. For these functions, the same argument applies for every complex $\lambda = a + ib$, using $e^{\lambda x} = e^{ax}(\cos bx + i \sin bx)$ and the elementary trigonometric derivative rules. Hence all scalars of the selected field occur, unlike the polynomial case.

**HW6 P3.**

**Idea.** The invertible map $S$ sends the equation for $R$ to the equation for $T$.

1. $Rv = \lambda v$ iff $S^{-1} T S v = \lambda v$, iff $T(Sv) = \lambda S v$ after multiplication by $S$.
2. Thus $v \in E(\lambda, R)$ iff $Sv \in E(\lambda, T)$, proving $E(\lambda, R) = S^{-1} E(\lambda, T)$.
3. Since $S$ is invertible, it sends nonzero vectors to nonzero vectors and back. Therefore one eigenspace is nonzero iff the other is, giving equal eigenvalues.

**HW6 P4.**

**Idea.** For two-by-two matrices, noninvertibility is exactly a zero elementary determinant.

1. $tI - A = \left(\begin{smallmatrix} t-a & -b \\ -c & t-d \end{smallmatrix}\right)$, so its determinant is $(t-a)(t-d) - bc = t^2 - (a+d)t + (ad - bc)$.
2. $\lambda$ is an eigenvalue iff $(\lambda I - A)v = 0$ for nonzero $v$, iff $\lambda I - A$ is not injective. Finite dimension makes this equivalent to noninvertibility.
3. Week 4's two-by-two criterion makes that equivalent to $\det(\lambda I - A) = 0$.

**HW6 P5: $A_1$.**

**Idea.** First specify the field, then solve the eigenvalue equation.

1. $\chi_{A_1}(t) = t^2 + 1$, so over $\mathbb{R}$ there are no eigenvalues and no eigenvector basis.
2. Over $\mathbb{C}$ the eigenvalues are $i, -i$. The equations $-y = \lambda x, x = \lambda y$ give $E(i, A_1) = \mathbb{C}(1, -i)$ and $E(-i, A_1) = \mathbb{C}(1, i)$.
3. The two eigenvalues are distinct, so their displayed eigenvectors are independent by Proposition 54 and form a basis of $\mathbb{C}^2$.

**HW6 P5: $A_2, A_3$.**

**Idea.** A root gives a candidate eigenvalue; the kernel dimension determines how many independent eigenvectors it supplies.

1. For $A_2$, $\chi(t) = (t-1)^2$ and $(A_2 - I)(x, y) = (y, 0)$. Thus the sole eigenvalue is 1, with $E(1, A_2) = \mathbb{R}e_1$. A one-dimensional eigenspace cannot supply a basis of $\mathbb{R}^2$.
2. For $A_3$, $\chi(t) = t^2 - t - 1$, with $\lambda_\pm = (1 \pm \sqrt{5})/2$. Solving gives $E(\lambda_\pm, A_3) = \mathbb{R}(1, \lambda_\pm)$.
3. These eigenvalues are distinct, so the two displayed eigenvectors form a basis of $\mathbb{R}^2$ by Proposition 54.

**HW6 P6 (optional).**

**Idea.** The first dependent iterate makes a finite initial span invariant.

1. A vector of $C(v, T)$ is a finite sum $\sum_{j=0}^{r} a_j T^j v$. Applying $T$ gives $\sum_{j=0}^{r} a_j T^{j+1} v$, again a finite combination of the iterates. Thus the cyclic subspace is invariant.
2. Let $n = \dim V$. Since $v \neq 0$ and the first $n+1$ iterates are dependent, there is a first index $m$ with $1 \leq m \leq n$ such that $T^m v \in \operatorname{span}(v, \ldots, T^{m-1} v)$.
3. The span $U$ of those first $m$ iterates is invariant: $T$ sends each generator except the last to the next, and sends the last into $U$ by the first-dependence relation.
4. Induction gives every later iterate in $U$, so $C(v, T) = U$, as required.

**HW6 P7 (optional).**

**Idea.** Stationarity forces equal coordinates, and total probability fixes their value.

1. Write $p = (a, b, c)^\mathsf{T}$. The first equation $a = (a+b)/2$ gives $a = b$; the third $c = (b+c)/2$ gives $c = b$.
2. The second equation $b = (a+c)/2$ then holds. Since $a + b + c = 1$, the probability vector is $p = (1/3, 1/3, 1/3)^\mathsf{T}$.
3. Thus if the chain starts uniformly over the three states, its distribution remains uniform after each transition.

**Exam practice, Problem 7.**

**Idea.** Invariance passes to annihilators because precomposition evaluates only on images that stay in the subspace.

1. If $v \in \ker T$, then $Tv = 0 \in \ker T$; if $w = Tv \in \operatorname{range} T$, then $Tw = T(Tv) \in \operatorname{range} T$. Thus both subspaces are invariant.
2. More generally, let $U$ be invariant and $\varphi \in U^0$. For each $u \in U$, $Tu \in U$, so $(T'\varphi)(u) = \varphi(Tu) = 0$.
3. Therefore $T'\varphi \in U^0$, proving annihilator invariance. Apply this to $U = \ker T$ and $U = \operatorname{range} T$.